

23 **C** $I = BIL \sin \theta = (4 \times 10^{-3})(3)(2) \sin 90^\circ = 240 \text{ } \mu\text{N}$

24 **A** A neutral point (i.e., a point of zero magnetic field) is possible at a position where the magnetic flux densities from 2 current sources are in opposite direction and of the same magnitude.
Using Right-Hand Grip Rule, this point can only be achieved at region 2.

25 **C** Current is able to flow both ways due to two diodes in opposite directions. One